SINGAPORE MANAGEMENT UNIVERSITY

OPIM 201 Operations Management

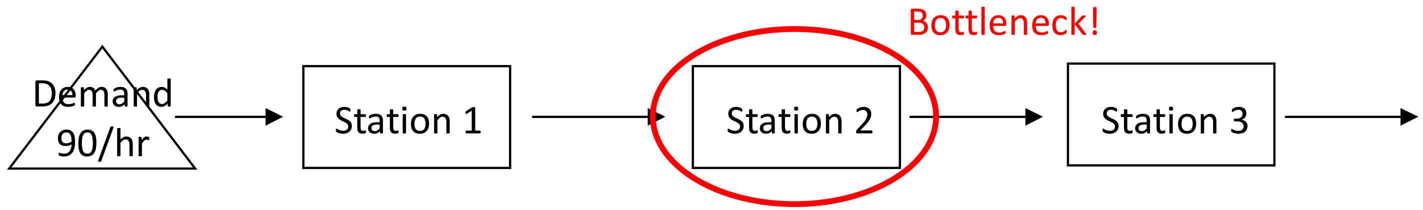
LINE BALANCING

Learning Objective

- Understand ***Line Balancing*** and how it can help improve process efficiency for line-flow processes
  - Line Balancing: allocating tasks so as to balance the workload throughout the process while meeting the target flow rate – for a line-flow process

Recall Subway Case: Hourly Demand of 90 Sandwiches

Ignore Task G, Toasting (oven)



	Worker 1	**Worker 2**	**Worker 3**
Activity Time	37 seconds/unit	46 seconds/unit	37 seconds/unit
Capacity	1/37 unit per second = 97.3 units/hour	1/46 unit per second = 78.3 units/hour	1/37 unit per second =97.3 units/hour
Process Capacity	Minimum {97.3, 78.3, 97.3 units/hour} =78.3 units/hour		
Flow rate	Demand = 90 units/hour (Target CT = $^{60\times 60}/_{90}$ = 40 seconds) Flow rate = Minimum {demand, process capacity} = 78.3 units/hour		
Utilization	$^{78.3}/_{97.3} = 80.5\%$	$^{78.3}/_{78.3} = 100\%$	$^{78.3}/_{97.3} = 80.5\%$

The process is _capacity_-constrained. (System CT (= 46 sec.) > Target CT (= 40 sec.))

Avg. labor utilization = $\frac{80.5+100+80.5}{3} \approx 87\%$

Labor cost per sandwich = $\frac{3\text{ workers}\times\$12/\text{hr}}{78.3\text{ units/hr}} = \$0.46/\text{unit}$

Hourly Demand: **90** Sandwiches

Station	Task	Seconds	3+ Support
Station 1	Greet Customer	4	1
	Take Order	5	
	Get Bread/Wrap/Salad Bowl	4	
	Cut Bread	3	
	Meat	12	
	Cheese	9	
	Toasting	30	Places / Pulls
Station 2	Onions	3	2
	Lettuce	3	
	Tomatoes	4	
	Cucumbers	5	
	Pickles	4	
	Green Peppers	4	
	Black Olives	3	
	Hot Peppers	2	
	Place Condiments	5	
	Wrap/Napkins & Bag	13	
Station 3	Offer Fresh Value Meal™	3	3
	Offer Cookies	14	
	Ring on Register	20	
	Total Seconds	120 - 150	
Support Position	Phone/Fax Orders		4
	Baking Bread/Cookies		
	Filling In on Line		
	Restocking Sandwich Unit		
	Cleaning Customer Area		

Station	Activity Time (s)
1	37
2	**46**
3	37

Flow Rate = **78.3** units/hr

Target CT = **40** seconds

How can one increase the production to meet demand?

- Further job specialization at Station 2 (need to add worker)
- Line (Re)Balancing
  - Can we arbitrarily move task(s) from bottleneck to non-bottleneck(s)?

Line Balancing Can Be Very Complex

BOEING 737 has almost 6 million parts

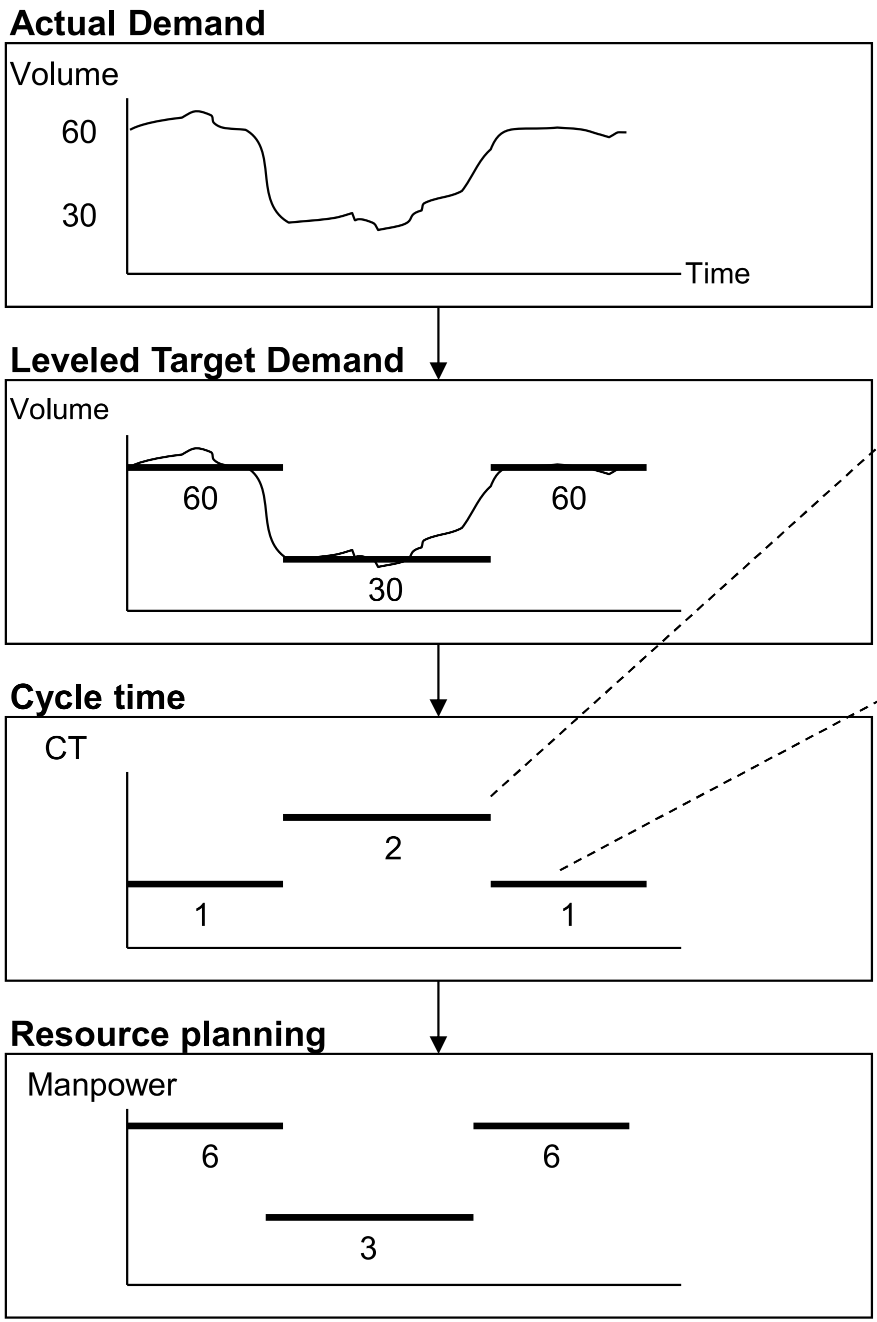
How do we perform line balancing or design a line?

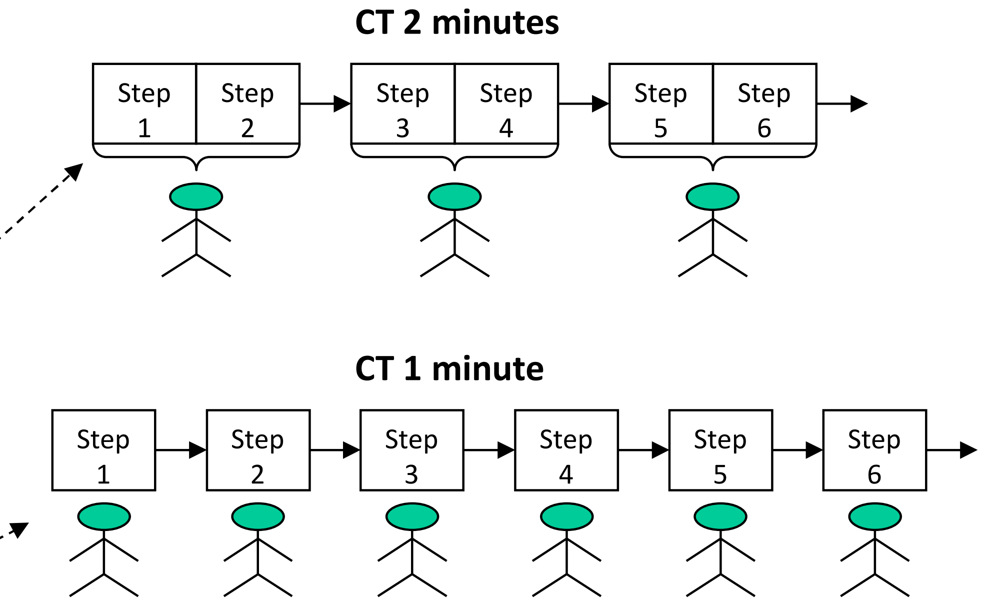
Ad hoc approach? Systematic approach?

Production Levelling

- **Production Levelling (Smoothing)**: Setting a constant production (or target demand) rate for a given period of time. Reasons:
  - Planning purposes (e.g. determining staffing plan)
    - Planning becomes too complex if actual demand variation is incorporated.
  - Production consistency and efficiency. (How?)
- Facilitated by Demand Levelling and Staffing to Demand
  - Demand levelling: deliberate influencing of demand to mitigate demand fluctuation.
  - Staffing to Demand: Adjusting capacity to meet demand.
    - **Line Balancing:** Allocate activities so as to balance the workload across the process while meeting the target demand.
      - Consider automating time-consuming activities, outsourcing activities, adding employees, overtime.
      - Consider Specialization: Breaking down task (especially bottleneck).
- **Capacity Flexibility** is crucial as target demand rate may vary across periods.
  - Need a long-term perspective – Cross-training, temporary or flexible workforce

Line Balancing and Staffing to Demand





Capacity flexibility

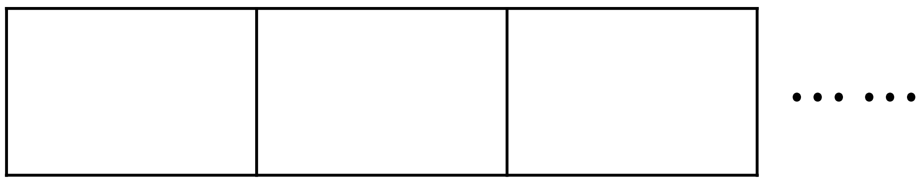
Ability to adjust to changing demands

Often implemented with temporary workers

Keeps average labor utilization high

Line Balancing – Problem Description

- Assign all tasks to a sequence of workstations.
- Objective:
  - Minimize number of workstations;
    - Maximize the overall labor/machine utilization for a given production rate/flow rate;
  - Minimize idle time across all workstations (a “good” balance).
- Constraints:
  - Target cycle time or flow rate requirement; (recall R = 1/CT)
  - Precedence relationship (specifies the order in which tasks must be performed).



Processing time at every station is no more than Target CT

Basic Line Balancing Vocabulary

- (Target) Cycle time, CT : the time between successive units coming off the end of the line.
- Task time, t_k : time it takes to complete task k
- A task is **eligible** to be included in a workstation, if all its preceding tasks have been included in the current or prior workstations.

Line Balancing Example – Huffy

Huffy assembles bicycles on a conveyor belt and operates one shift (8 hours) per day. Currently its daily forecast is 220 bicycles and its daily production plan is 225 bicycles. Perform a line balance of the assembly line that achieves the target production rate with the following tasks to be performed on the line:

Task	Time (sec.)	Predecessor
1	60	None
2	45	1
3	35	1
4	92	3
5	55	3
6	70	2, 4
7	30	6
8	25	5, 6
9	65	7, 8
Total	477	

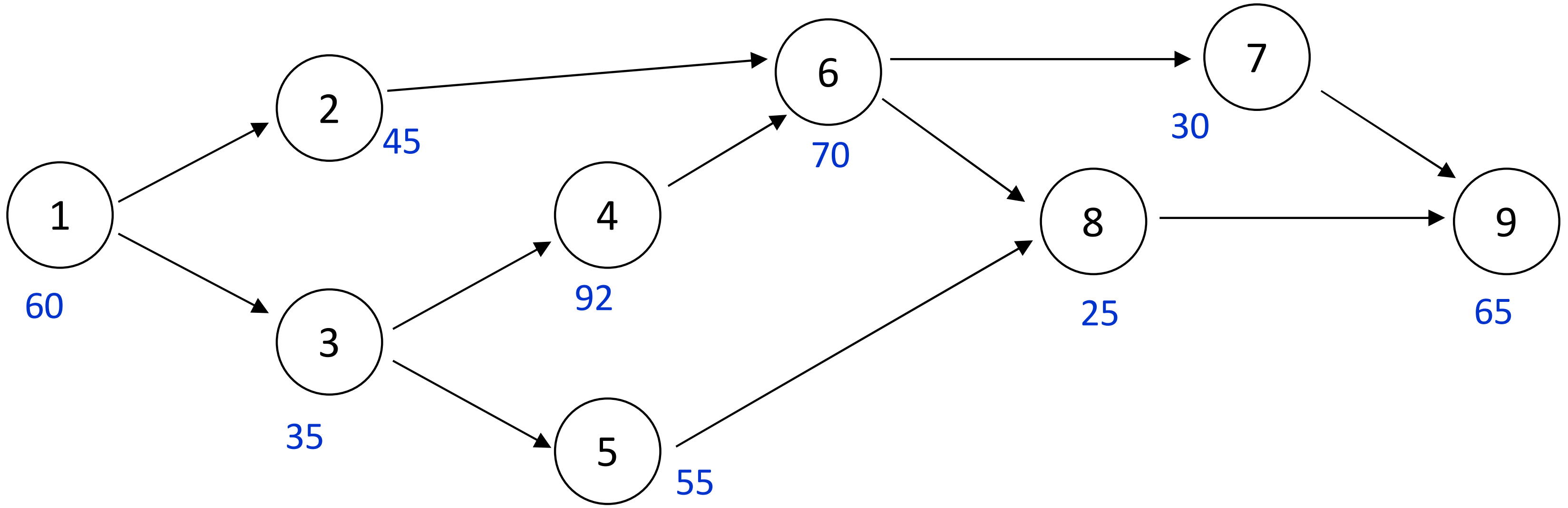
What is the target production rate?

The Precedence Diagram

The Precedence Diagram depicts the Precedence Relationship.

Task	Predecessors
1	None
2	1
3	1
4	3
5	3

Task	Predecessors
6	2, 4
7	6
8	5, 6
9	7, 8



Compute Target CT

Target CT = 1/225 day per unit (R = 225 units/day)

= (8 hr/day)(60 min/hr)(60 sec/min) / 225 units

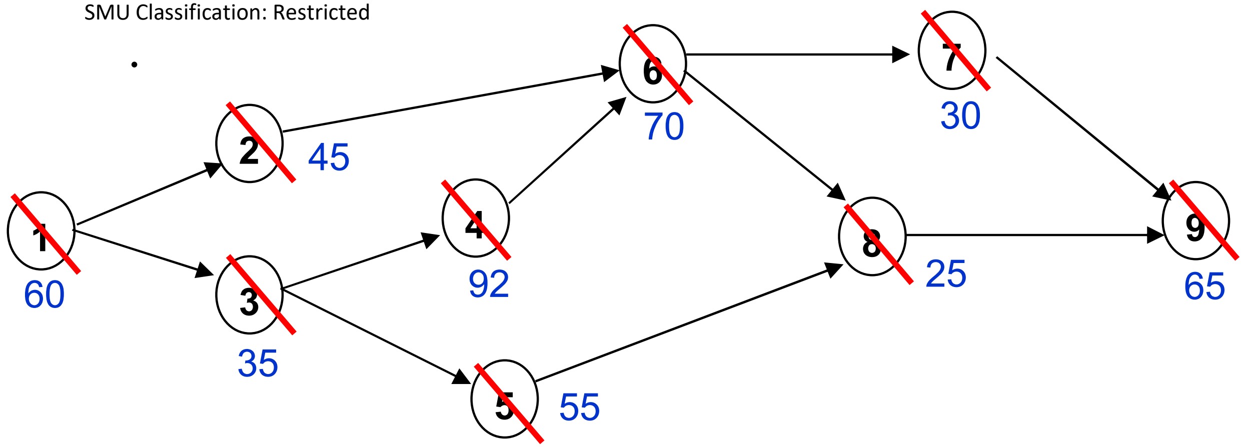
= 128 sec. per unit

Two Heuristics (Priority Rules) for Line Balancing:

- Primary rules
  - Longest Task Time (LTT) rule
    - Assign tasks (that fit within the remaining time constraint) in order of the longest task time
  - Largest Number of Followers (LNF) rule
    - Assign tasks (that fit within the remaining time constraint) in order of the largest number of following tasks
- Tie breaker
  - When applying a priority rule, if two or more tasks are tied with same priority value, then use the other priority rule as a tie breaker

The above two are heuristics; they don't necessarily find the minimum number of workstations for a target productivity.

LTT Heuristic



Workstation	Time Remaining	Eligible	Assign	Idle
1	128 68	1 2, 3	1 2	23
2	128 93	3 4, 5	3 4	1
3	128 58	5, 6 5, 7	6 5	3
Optimal? 4	128 98 73	7, 8 8 9	7 8 9	8
Total				35

Theoretical Minimum Number of Workstations

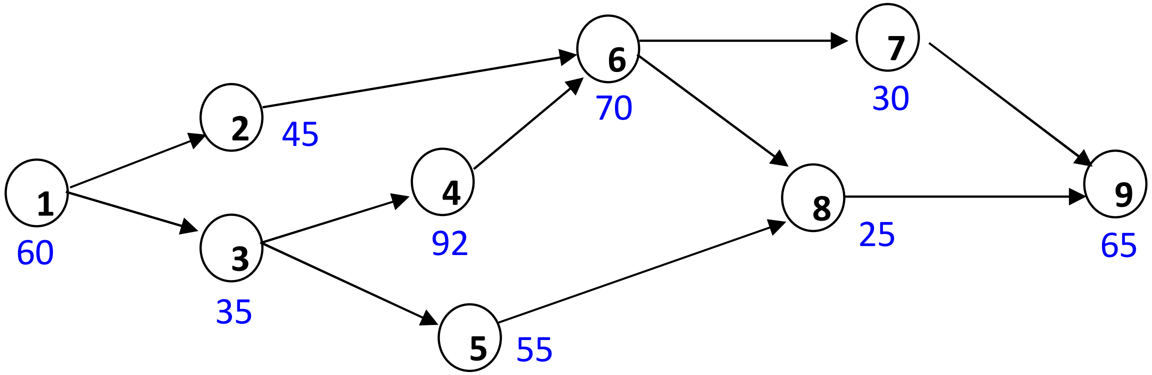
$N_{min} = \Sigma t_k / CT$ (t_k = task time of task k)

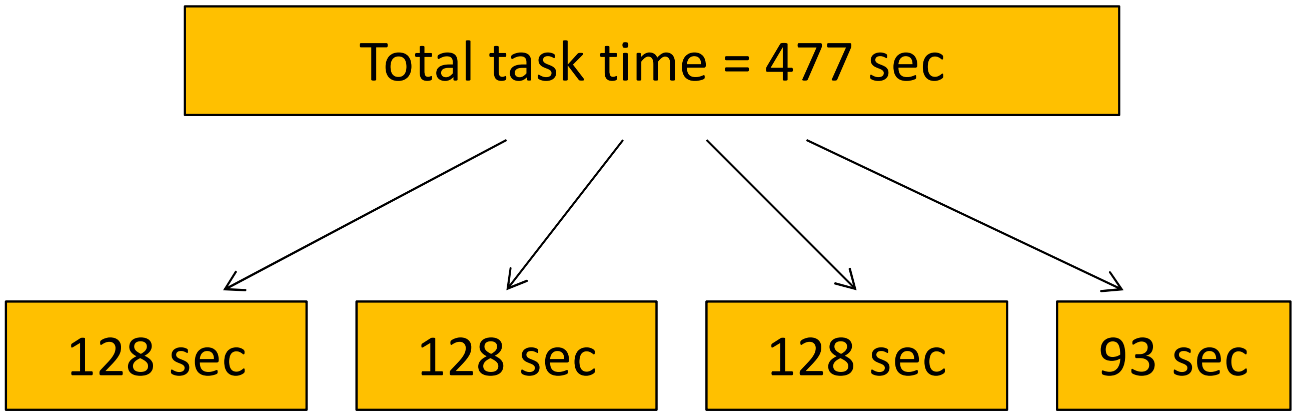
$= 477$ sec. / 128 sec.

$= 3.73$ or 4 workstations

Can be interpreted as Target Manpower

Can't do better than 4 stations!





LTT Heuristic (cont'd)

- Rationale of LTT?
- Comment on the task allocation.
- What do you think of the idle time?

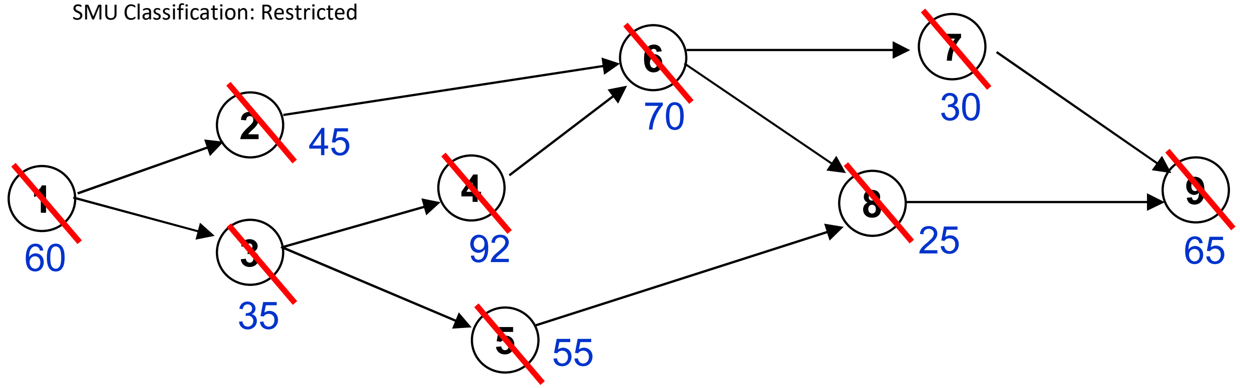
Idle time in all stations!
Reduce CT & idle time?

Workstation	Time Remaining	Eligible	Assign	Idle
1	128 68	1 2, 3	1 2	23
2	128 93	3 4, 5	3 4	1
3	128 58	5, 6 5, 7	6 5	3
4	128 98 73	7, 8 8 9	7 8 9	8
Total				35

Addressing Idle Time …

- Make workers spend idle time on quality control/background processing
- Cross-trained workers to help busy workers
- Redesign process

LNF Heuristic



- LNF inferior?
- Rationale of LNF rule?

Workstation	Time Remaining	Eligible	Assign	Idle
1	128 68	1 2, 3	1 3	33
2	128	2, 4, 5	4	36
3	128 83	2, 5 5, 6	2 6	13
4	128 73 43	5, 7 7, 8 8	5 7 8	18
5	128	9	9	63
Total				163

Efficiency of the Line Flow for Huffy Under LNF Heuristic

Total workload or busy time or what actually gets done

- **Efficiency** $= \dfrac{\Sigma t_k}{N_a \times CT}$

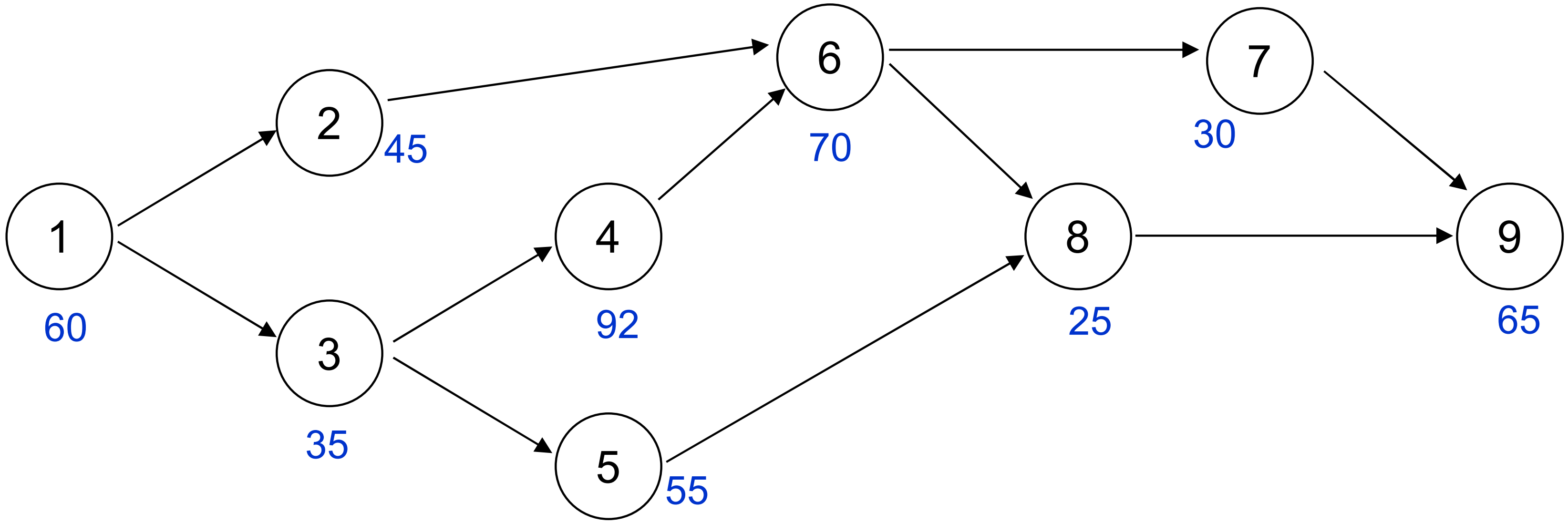
Maximum workload allowed or available time or what theoretically can get done

$$= \frac{477}{5 \times 128}$$

$$= 0.745 \text{ or } 74.5\%$$

- N_a = actual number of workstations
- Efficiency can be interpreted as the average utilization of the line; hence it's also equal to the average utilization of all workstations.

Maximum Production Rate?



The *minimum* cycle time of the line, and therefore, the maximum production rate is determined by task __4__ (bottleneck task)

Assume operating 1 shift (8 hrs) per day.

Max Production = $^{1}/_{92}$ units per sec

$\approx$ 313 units per day

Note that target production may not equal maximum production.

Remarks

- The LTT and LNF are heuristics; they don't necessarily find the OPTIMAL solution
- However, if in a solution, the number of stations is equal to the *theoretical minimum number of stations*, then the solution is optimal and efficiency is optimized
- Optimal number of stations may be more than theoretical minimum due to
  - Discrete values of task time
  - Precedence relationship
- Other heuristics?

Basic Line Balancing Vocabulary

- (Target) Cycle time, CT : the time between successive units coming off the end of the line.
- Task time, t_k : time it takes to complete task k
- A task is **eligible** to be included in a workstation, if all its preceding tasks have been included in the current or prior workstations.
- Efficiency = $\frac{\text{total task time}}{N_a \times \text{cycle time}}$, where N_a = actual number of workstations
- Theoretical minimum number of stations, N_{min}: The theoretical minimum number of stations required to achieve the target flow rate or CT.
  - Calculate the ratio of total task time to the cycle time, $\frac{\text{total task time}}{\text{cycle time}}$, and round it up.
  - Optimal number of stations required could be more due to discrete values of task times, precedence relationships and other restrictions.

General Procedure for Line Balancing

- **Step 1**
  - Create a new station numbered k with station time = 0
- **Step 2**
  - List all eligible tasks. If there are no eligible tasks, stop (final cycle time = largest station time, number of workstations = current value of k).
- **Step 3**
  - Select the first eligible task according to a specified priority rule. If (task time + current station time) $\leq$ cycle time, go to step 5. Otherwise, go to step 4.
- **Step 4**
  - Ignore this eligible task. If there are more eligible tasks, return to step 3 otherwise increase k by 1 and return to step 1.
- **Step 5**
  - Remove the task from the eligible list, add it to the current station. Increment current station time by the task time, find new eligible tasks and return to step 2.